Krishna Karthikeyan

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The Hiring Team
The Francis Crick Institute
1 Midland Road
London NW1 1AT
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Dear Francis Crick Hiring Team,

I am Krishna Karthikeyan, a final-year B.Tech student in Biological Engineering at IIT Madras, applying for the Junior Data Scientist position at the Francis Crick Institute. My academic and research experience spans computational biology, genomics, transcriptomics, machine learning, and bioinformatics, and I am particularly interested in applying quantitative approaches to questions in cancer biology.

Through my coursework and projects, I have developed experience using Python and R to analyse biological datasets and have built a strong theoretical foundation in genomics, systems biology, statistics, and machine learning. In a recent pan-cancer transcriptomics project, I analysed TCGA RNA-seq data across 16 cancer types using dimensionality reduction, differential expression, pathway enrichment, and classification methods, including analysis of clear-cell renal-cell carcinoma. This strengthened my interest in understanding how molecular heterogeneity influences disease progression and treatment response.

During my internship at GenePath Diagnostics in Pune, I worked on metagenomic and amplicon-sequencing analysis and developed an end-to-end pipeline for processing sequencing data and assigning taxonomy. One experience that I am particularly proud of was when I questioned results

produced by an existing Illumina BaseSpace workflow. Rather than accepting the software output at face value, I investigated the underlying analysis, compared alternative approaches, and obtained results that were more biologically and mathematically consistent. This reinforced the importance of critically evaluating computational outputs rather than treating analytical tools as black boxes. It also gave me perspective on drawing appropriate conclusions from underlying data.

I am particularly interested in the Cancer Dynamics Laboratory because of its work on cancer evolution, tumour heterogeneity, and the integration of genomic and transcriptomic data to understand disease progression and treatment response. The opportunity to contribute to these questions using computational approaches is closely aligned with the direction in which I want to develop my career. I have attached the report of a related course project I have undertaken in this field.

IIT Madras permits students to undertake a six-month research or industry placement during the final semester of the B.Tech programme, and I would be very interested in undertaking this placement at the Crick Institute. After graduation, I intend to work in computational biology or biomedical data science for two years before pursuing postgraduate study. I am therefore looking for an environment in which I can contribute meaningfully to ongoing research while developing the depth of scientific and computational experience required for a long-term career in this field.

I believe my combination of biological training, quantitative problem-solving, and willingness to critically interrogate results would allow me to contribute effectively to the team. I would welcome the opportunity to discuss my suitability for the role further.

Thank you for considering my application.

Yours faithfully,

Krishna Karthikeyan